Ice-column damage evolution from van der Veen crevasse mechanics

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Abstract

Surface and basal crevasses determine whether an ice column retains a connected load-bearing ligament. The van der Veen models predict whether an existing crevasse propagates or arrests, but they do not specify a propagation rate. This distinction prevents their direct use as a prognostic damage equation. Here, the fractured thickness fraction is written as $D = D_{\mathrm{s}} + D_{\mathrm{b}}$, and the van der Veen stress-intensity factors are retained as the fracture calculation. A Freund-style elastodynamic relation then converts fracture excess into crack-tip speed. The resulting law predicts the same arrest depths as the original criterion and declares full-thickness failure when $D = 1$. The speed approaches a finite limiting value as the stress-intensity factor increases. This value is the Rayleigh wave speed for ideal homogeneous ice, but may require calibration when the model represents unresolved crevasse-scale processes.

1 Mechanical question

Crevasse-depth calculations provide a geometric measure of ice damage, but a static depth is not an evolution law. In particular, the linear elastic fracture mechanics (LEFM) models of van der Veen (1998b) and van der Veen (1998a) calculate the mode-I stress intensity factor at an assumed crack tip. They predict propagation when this factor exceeds the fracture toughness of ice. They do not calculate $\mathrm{D}D/\mathrm{D}t$.

This note makes the smallest additional assumption needed to obtain that rate. The calculation is separated into two parts. First, van der Veen mechanics determines whether each crack tip is above or below its propagation threshold. Second, a Freund-style kinetic function maps the threshold excess to a crack-tip speed. A reduced zero-stress version is also given because it leads to an algebraic right-hand side suitable for a depth-integrated ice model. The reduced version is a separate approximation and is not the van der Veen LEFM calculation.

2 Geometry, boundary conditions, and assumptions

Consider an ice column of thickness H measured in meters. A surface crevasse penetrates downward a distance d_{s}, and a basal crevasse penetrates upward a distance d_{b}. The state variables are

$$D_{\mathrm{s}} = \frac{d_{\mathrm{s}}}{H}, \qquad D_{\mathrm{b}} = \frac{d_{\mathrm{b}}}{H}, \qquad D = D_{\mathrm{s}} + D_{\mathrm{b}}. \tag{1}$$

The model applies while $0 \le D < 1$. Full-thickness connection occurs at the first-hitting time

$$t_{\mathrm{f}} = \min\{t : D(t) = 1\}. \tag{2}$$

At t_{f}, the column is declared fractured through its thickness; the damage equation is not continued beyond this event.

Table 1: Notation. Subscripts s, b, i, and m denote surface, basal, ice, and surface meltwater quantities, respectively.

Symbol	Meaning and SI units	Symbol	Meaning and SI units
H	Ice thickness (m)	$d_\mathrm{s}, d_\mathrm{b}$	Surface and basal penetration depths (m)
$D_\mathrm{s}, D_\mathrm{b}$	Surface and basal fractured fractions (dimensionless)	D	Total fractured fraction, $D_\mathrm{s} + D_\mathrm{b}$ (dimensionless)
$\boldsymbol{u}$	Horizontal ice velocity ($\mathrm{m\,s^{-1}}$)	$\mathcal{D}_t$	Material derivative ($\mathrm{s^{-1}}$ when applied to D)
R_n	Resistive stress normal to a crevasse (Pa)	τ_{nn}, τ_{tt}	Deviatoric stresses normal and parallel to a crevasse (Pa)
ρ_i	Ice density ($\mathrm{kg\,m^{-3}}$)	$\rho_\mathrm{m}, \rho_\mathrm{b}$	Surface and basal water densities ($\mathrm{kg\,m^{-3}}$)
g	Gravitational acceleration ($\mathrm{m\,s^{-2}}$)	$a_\mathrm{s}, a_\mathrm{b}$	LEFM surface and basal crack lengths (m)
h_s	Surface water-column height (m)	δ_s	Dry length above water in a surface crack (m)
p_b^0	Basal water pressure at the crack mouth (Pa)	$K_\mathrm{I}^\mathrm{s}, K_\mathrm{I}^\mathrm{b}$	Mode-I stress intensity factors ($\mathrm{Pa\,m^{1/2}}$)
K_Ic	Ice fracture toughness ($\mathrm{Pa\,m^{1/2}}$)	F, G	Finite-slab weight functions (dimensionless)
λ, γ	Normalized crack length and integration coordinate (dimensionless)	$v_\mathrm{s}, v_\mathrm{b}$	Crack-speed scales ($\mathrm{m\,s^{-1}}$)
$\mathcal{F}_j, c_\mathrm{R}$	Kinetic function (dimensionless) and Rayleigh wave speed ($\mathrm{m\,s^{-1}}$)	Σ	Resistive stress normalized by ice overburden (dimensionless)
W_s	Surface water-column fraction (dimensionless)	$r_\mathrm{s}, r_\mathrm{b}$	Water-to-ice density ratios (dimensionless)
H_ab	Thickness above basal water-pressure support (m)	A_b	H_ab/H (dimensionless)
$\Psi_\mathrm{s}, \Psi_\mathrm{b}$	Reduced normalized opening stresses (dimensionless)	t_f	First time at which $D = 1$ (s)

Table 1 summarizes the notation used below.

The calculation makes five assumptions.

1. Surface and basal crevasses are pre-existing mode-I cracks. The model predicts propagation, not nucleation, and does not include healing.
2. The tensile resistive stress R_n, measured in pascals, is uniform with depth. Ice overburden and water pressure are hydrostatic.
3. Surface water-column height and basal water pressure are prescribed by observations or a separate hydrology model.
4. Surface and basal cracks do not mechanically interact before $D = 1$. This assumption isolates their separate contributions but becomes least accurate when the remaining ligament is short.
5. The dimensionless fractions are advected with the ice. Crack-tip speed is measured relative to affine vertical deformation of the column.

With horizontal ice velocity $\boldsymbol{u}$, the material derivative is

$$\mathcal{D}_t(\,\cdot\,) = \frac{\partial(\,\cdot\,)}{\partial t} + \boldsymbol{u} \cdot \nabla(\,\cdot\,). \tag{3}$$

The resistive stress should be calculated normal to the crevasse. In a three-dimensional shallow model this gives $R_n = 2\tau_{nn} + \tau_{tt}$, where t is parallel to the crevasse; using τ_{nn} alone is not mechanically equivalent (Reynolds et al., 2025).

3 van der Veen fracture calculation

The van der Veen criterion is

$$K_{\mathrm{I}}^{j} > K_{\mathrm{Ic}}, \qquad j \in \{\mathrm{s}, \mathrm{b}\}, \tag{4}$$

where K_{I}^{j} and K_{Ic} have units of $\mathrm{Pa\,m^{1/2}}$. A crack propagates while equation (4) holds and arrests at the first stable depth where $K_{\mathrm{I}}^{j} = K_{\mathrm{Ic}}$.

For a surface crack, let $a_{\mathrm{s}} = HD_{\mathrm{s}}$, let y increase downward from the ice surface, and let δ_{s} be the dry crack length above the water surface. The water-column height is $h_{\mathrm{s}} = \langle a_{\mathrm{s}} - \delta_{\mathrm{s}} \rangle_{+}$. The surface stress intensity factor is

$$\begin{aligned} K_{\mathrm{I}}^{\mathrm{s}} = F(\lambda_{\mathrm{s}}) R_n \sqrt{\pi a_{\mathrm{s}}} - \frac{2\rho_{\mathrm{i}} g}{\sqrt{\pi a_{\mathrm{s}}}} \int_0^{a_{\mathrm{s}}} y G\left(\lambda_{\mathrm{s}}, \frac{y}{a_{\mathrm{s}}}\right) \mathrm{d}y \\ + \frac{2\rho_{\mathrm{m}} g}{\sqrt{\pi a_{\mathrm{s}}}} \int_0^{a_{\mathrm{s}}} \langle y - \delta_{\mathrm{s}} \rangle_{+} G\left(\lambda_{\mathrm{s}}, \frac{y}{a_{\mathrm{s}}}\right) \mathrm{d}y, \qquad \lambda_{\mathrm{s}} = \frac{a_{\mathrm{s}}}{H}. \end{aligned} \tag{5}$$

The three terms are opening by resistive stress, closure by ice overburden, and opening by meltwater pressure.

For a basal crack, let $a_{\mathrm{b}} = HD_{\mathrm{b}}$, let y increase upward from the bed, let p_{b}^{0} be water pressure at the crack mouth, and let ρ_{b} be the density of water in the crack. The basal stress intensity factor is

$$K_{\mathrm{I}}^{\mathrm{b}} = \frac{2}{\sqrt{\pi a_{\mathrm{b}}}} \int_0^{a_{\mathrm{b}}} \left\{ R_n - \rho_{\mathrm{i}} g (H - y) + \langle p_{\mathrm{b}}^{0} - \rho_{\mathrm{b}} g y \rangle_{+} \right\} G\left(\lambda_{\mathrm{b}}, \frac{y}{a_{\mathrm{b}}}\right) \mathrm{d}y, \qquad \lambda_{\mathrm{b}} = \frac{a_{\mathrm{b}}}{H}. \tag{6}$$

Here ice overburden closes the crack, whereas basal water pressure and tensile resistive stress open it. Appendix A gives the finite-slab weight functions F and G.

LEFM requires a finite initial flaw because K_I^j is not defined at zero crack length. The initial condition must therefore specify $D_j(0) = D_{j0} > 0$. In addition, the finite-slab weight functions assume a rectangular body with a single edge crack. Grounded basal boundary conditions can change the calculated stress intensity factor, particularly for deep cracks (Jiménez and Duddu, 2018).

4 Damage evolution

The static criterion fixes the sign of the crack speed but not its magnitude. Write the unknown dimensionless kinetic relation as

$$\mathcal{F}_j(q) = 0 \quad \text{for } q \le 1, \qquad \mathcal{F}_j(q) > 0 \quad \text{for } q > 1. \tag{7}$$

If v_j is a crack-speed scale in $\mathrm{m\,s^{-1}}$, a local rate law consistent with the criterion is

$$\mathcal{D}_t D_\mathrm{s} = \frac{v_\mathrm{s}}{H}\mathcal{F}_\mathrm{s}\left(\frac{K_\mathrm{I}^\mathrm{s}}{K_\mathrm{Ic}}\right), \tag{8}$$

$$\mathcal{D}_t D_\mathrm{b} = \frac{v_\mathrm{b}}{H}\mathcal{F}_\mathrm{b}\left(\frac{K_\mathrm{I}^\mathrm{b}}{K_\mathrm{Ic}}\right). \tag{9}$$

These equations have units of inverse time. They also retain the original arrest condition because both rates vanish when $K_\mathrm{I}^j \le K_\mathrm{Ic}$.

A Freund-style closure follows from the elastodynamic factorization of the moving-tip stress-intensity factor (Freund, 1990). In the form used for ice-shelf rifts by Lipovsky (2018), it is

$$\mathcal{F}_j(q) = \mathcal{F}(q) = \begin{cases} 0, & q < 1, \\ 1 - q^{-2}, & q \ge 1. \end{cases} \tag{10}$$

Here K_I^j is the stress-intensity factor for a stationary crack of the instantaneous length and $q = K_\mathrm{I}^j / K_\mathrm{Ic}$. The function vanishes at the propagation threshold and approaches unity as q increases. Near the threshold, $\mathcal{F}(q) = 2(q-1) + \mathcal{O}\{(q-1)^2\}$. Setting $v_j = c_\mathrm{R}$, where c_R is the Rayleigh wave speed in ice, recovers the ideal homogeneous-material relation. We retain v_j as an effective limiting speed so that unresolved dissipation can be calibrated. This relation is not derived by van der Veen. With equation (10), the requested right-hand side is

$$\boxed{\mathcal{D}_t D = \frac{v_\mathrm{s}}{H}\mathcal{F}\left(\frac{K_\mathrm{I}^\mathrm{s}(D_\mathrm{s})}{K_\mathrm{Ic}}\right) + \frac{v_\mathrm{b}}{H}\mathcal{F}\left(\frac{K_\mathrm{I}^\mathrm{b}(D_\mathrm{b})}{K_\mathrm{Ic}}\right), \qquad D < 1.} \tag{11}$$

Equation (11), together with the two component equations, is the prognostic van der Veen damage model. The calculation retains the original propagation threshold; the new parameters v_s and v_b set only the transient rate.

5 Reduced algebraic form

Some depth-integrated models do not evaluate stress-intensity factors. A separate zero-stress approximation places each crack tip where its net opening stress vanishes (Nye, 1955; Nick et al., 2010). Define

$$\Sigma = \frac{R_n}{\rho_\mathrm{i} g H}, \qquad W_\mathrm{s} = \frac{h_\mathrm{s}}{H}, \qquad r_\mathrm{s} = \frac{\rho_\mathrm{m}}{\rho_\mathrm{i}}, \qquad r_\mathrm{b} = \frac{\rho_\mathrm{b}}{\rho_\mathrm{i}}, \tag{12}$$

and

$$A_\mathrm{b} = \frac{H_\mathrm{ab}}{H}, \qquad H_\mathrm{ab} = H - \frac{p_\mathrm{b}^0}{\rho_\mathrm{i} g}. \tag{13}$$

Here Σ is tensile stress normalized by ice overburden, W_s is fractional surface-water depth, and A_b measures the ice thickness not supported by basal water pressure. Assuming water reaches the basal crack tip, the normalized opening stresses are

$$\Psi_\mathrm{s} = \Sigma + r_\mathrm{s} W_\mathrm{s} - D_\mathrm{s}, \tag{14}$$

$$\Psi_\mathrm{b} = \Sigma - A_\mathrm{b} - (r_\mathrm{b} - 1) D_\mathrm{b}. \tag{15}$$

Positive Ψ_j opens the corresponding crack. This approximation does not evaluate $K_\mathrm{I}^j / K_\mathrm{Ic}$, so the Freund relation cannot be applied without an additional mapping between opening stress and stress intensity. Retaining a separate linear algebraic closure gives

$$\boxed{\mathcal{D}_t D = \frac{v_\mathrm{s}}{H} \langle \Sigma + r_\mathrm{s} W_\mathrm{s} - D_\mathrm{s} \rangle_+ + \frac{v_\mathrm{b}}{H} \langle \Sigma - A_\mathrm{b} - (r_\mathrm{b} - 1) D_\mathrm{b} \rangle_+ , \qquad D < 1.} \tag{16}$$

The two component equations are obtained by retaining the corresponding term on the right-hand side. In this reduced closure, v_j is a rate scale and does not impose the Freund speed limit.

The static limits provide a direct check. A dry surface crack arrests at $D_\mathrm{s}^\star = \Sigma$. A water-filled basal crack arrests at

$$D_\mathrm{b}^\star = \frac{1}{r_\mathrm{b} - 1} \langle \Sigma - A_\mathrm{b} \rangle_+ , \tag{17}$$

provided $\rho_\mathrm{b} > \rho_\mathrm{i}$. These are the standard zero-stress crevasse depths. In contrast, a fully water-filled surface crack has $W_\mathrm{s} = D_\mathrm{s}$, and therefore

$$\Psi_\mathrm{s} = \Sigma + (r_\mathrm{s} - 1) D_\mathrm{s}. \tag{18}$$

Freshwater is denser than ice, so $r_\mathrm{s} > 1$. Once this crack propagates, equation (18) does not admit a positive arrest depth; the reduced model predicts full-thickness hydrofracture if the water column remains supplied.

6 Predictions and limitations

No observations are analyzed in this note. The model nevertheless makes three testable predictions.

First, measured arrest depths should satisfy $K_\mathrm{I}^j = K_\mathrm{Ic}$ when the stress, thickness, and water-pressure boundary conditions are inserted into equations (5) and (6). The reduced model instead predicts the algebraic depths above. Radar measurements of basal crevasse height and repeated surface topography could distinguish these calculations where the required boundary conditions are known.

Second, if v_j is constant, the Freund-style kinetics predicts

$$\frac{H \mathcal{D}_t D_j}{1 - (K_\mathrm{Ic}/K_\mathrm{I}^j)^2} = v_j, \qquad K_\mathrm{I}^j > K_\mathrm{Ic}, \tag{19}$$

during propagation. Thus repeated measurements of crevasse depth should collapse to a constant speed for a given crack family. A systematic dependence of the left-hand side on fracture excess would reject the Freund-style closure while leaving the van der Veen threshold calculation intact.

Third, for $0 < \Sigma < 1$, the reduced model predicts that a dry surface crack arrests at $D_\mathrm{s} = \Sigma$, whereas a fully water-filled surface crack propagates through the thickness under the same tensile stress. This prediction is conditional on continued water supply. A hydrology model is required to determine whether that boundary condition persists.

The most important uncertainty is whether the Freund-style kinetic function, which describes rapid elastodynamic crack growth in a homogeneous elastic solid, applies at the scale of an ice-column model. Crack interaction, firn structure, spatially varying fracture toughness, and the geometric dependence of the weight functions are also omitted. These omissions affect the transient speed and may shift the arrest depth. They should be added only when observations require them.

7 Conclusion

This note rewrites the van der Veen propagation criterion as an evolution law for the fractured thickness fraction. The central result is equation (11): surface and basal damage grow only when their stress-intensity factors exceed ice fracture toughness, and their rates add until $D = 1$. The threshold and arrest calculation come from van der Veen mechanics. The primary LEFM model uses a Freund-style relation between fracture excess and crack speed; the reduced zero-stress model retains a separate linear closure. Time-resolved measurements of crevasse growth could test these relations or replace them without changing the underlying fracture calculation.

A Finite-slab weight functions

The van der Veen finite-slab functions, as reproduced by Jiménez and Duddu (2018), are

$$F(\lambda) = 1.12 - 0.23\lambda + 10.55\lambda^2 - 21.72\lambda^3 + 30.39\lambda^4, \tag{20}$$

and

$$\begin{aligned} G(\lambda,\gamma) = {} & \frac{3.52(1-\gamma)}{(1-\lambda)^{3/2}} - \frac{4.35 - 5.28\gamma}{(1-\lambda)^{1/2}} \\ & + \left[\frac{1.30 - 0.30\gamma^{3/2}}{(1-\gamma^2)^{1/2}} + 0.83 - 1.76\gamma\right]\left[1 - (1-\gamma)\lambda\right], \end{aligned} \tag{21}$$

where $\lambda = a/H$ and $\gamma = y/a$. The crack-tip singularity is integrable. Numerical quadrature should stop short of $\lambda = 1$, where the ice column has already met the full-thickness failure condition.

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